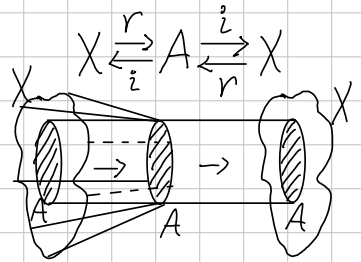


Def. Let X be a space, and $A \subset X$ be a subspace with inclusion $i: A \hookrightarrow X$.
 A continuous map $r: X \rightarrow A$ ($\subset X$) is called a retraction if $r \circ i = \text{id}_A$

In the Algebraic Topology, we say that for $A \subset X$,
 X deformation retract if there is a retraction $r: X \rightarrow A$ such that $i \circ r \simeq \text{id}_X$



Thm. Let $A \subset X$, $i: A \hookrightarrow X$ be an inclusion, and $f: X \rightarrow A$ be a continuous. Suppose that $i \circ f \simeq \text{id}_X$ via the Homotopy $H: X \times I \rightarrow X$.

Prop. The Möbius band M is Homotopy equivalent to the cylinder $S^1 \times I$.